JORDAN
INVESTMENT COMMISSION
الأردن
هيئة الاستثمار

A pre-feasibility study for the establishment of a factory for the manufacturing of pickles in Mafraq

2017

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1. Executive Summary

The process of pickling vegetables is one of the oldest means used by humans in keeping the surplus of food at the household level, effective vaible way to preserve vegetables from damage and corruption for a long time. Pickles are a popular product that is so desirable. It is one of the most demanded food, which are considered as appetizers, and have nutritional value. The pickling industry has been transformed from a process that was practiced at the domestic level into a large-scale commercial industry. The manufacturers have mastered this industry and introduced improvements and modifications in the manufacturing process andpreseving in Vinegar, or in the process of filling them in attractive packaging, or the methods of mixing different vegetables with each other as well as add spices to pickles.

The idea of the project is to produce different kinds of pickles such as cucumbers, olive pickles (black and green), peppers, etc., where vegetables are collected from farms and then cleaned and stored until they are pickled.

The idea was adopted because of the availability of manpower in the province, along with high profitability of such projects.

Table 1 Economic Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	670,264
Internal Rate of Return (IRR)	%22.45
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	473,539
Job Opportunities provided by the Project	18
Total Revenues (For 10 Years)	5,940,000
Total Operating Expenses (For 10 years)	3,505,866
Total Net Profit (For 10 years)	766,002

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

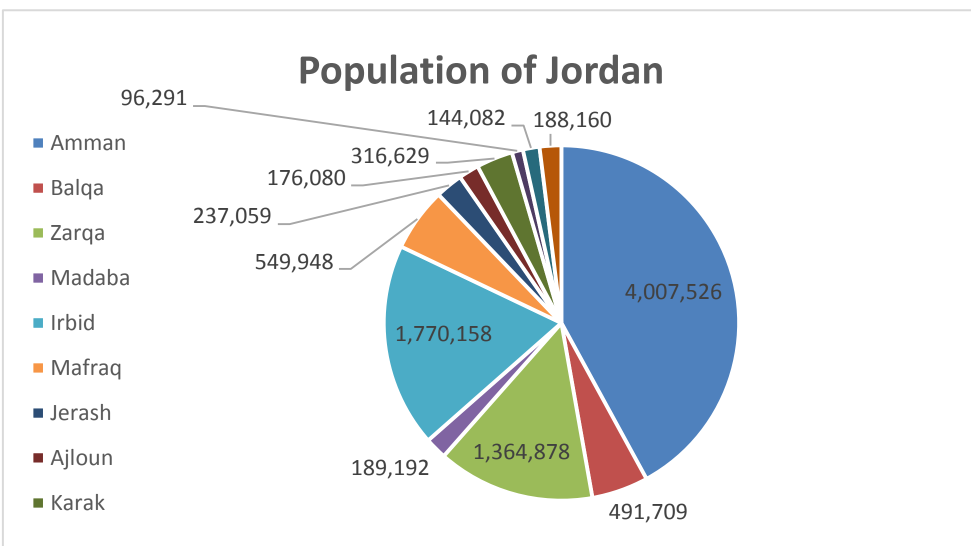
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

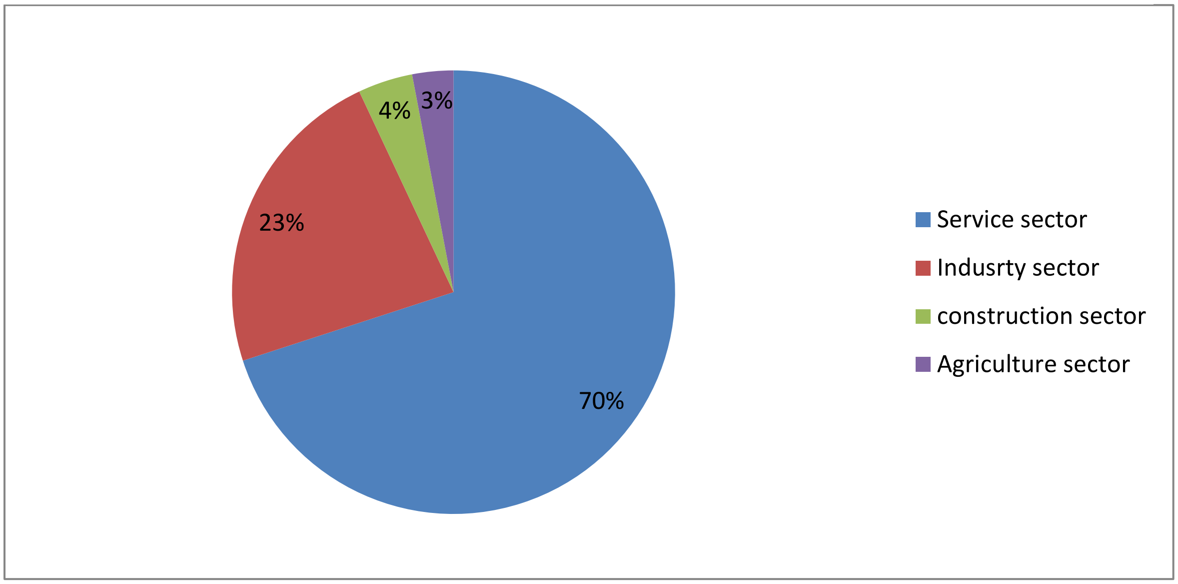
Department of Statistics (DOS), 2015.

Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Mafraq:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 Payable Income Tax and Tax Exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Mafraq governorate is classified as category C where investors get 60% tax exemption on income tax for 20 years from the date of investment.

2. Market Analysis

Project Description

The idea of the proposed project is to establish a factory that produce various types of pickles in Al Mafraq and get advantage of the opening of the King Hussein Bin Talal Development area, which includes many privileges for investors.

The idea of the project is to produce different kinds of pickles such as cucumbers, olive pickles (black and green), peppers, etc., where vegetables are collected from farms and then cleaned and stored until they are picked.

Project Objectives

• Achieve a profitable return to the owners.

• Provide job opportunities for people in the region and reduce unemployment.

• Promoting agricultural investment in the region.

• Contribute in the marketing of surplus vegetables and pickles to meet the needs of the local market.

Proposed Product

The proposed project is producing olives, cucumbers, and peppers pickles, where they are filled and packed in special packaging.

Target Segmants

- The project will target the local market in the following sectors:
- Restaurants, hotels, supermarkets, shops, dairy and cheese shops.

Market Size

It is difficult to count the quantities produced or even the factories that carry out this profession because most of them are in homes. However, the previous consumption of pickles can be estimated according to the following data and assumptions:

- The population of Jordan for 2016 was 9,814,959 and the average number of family members was 5.8. Which means that the number of households in Jordan reached about 1.7 million families.

- (20%) of the households do not consume pickles. Thus, the number of households using pickles with food is about 1.36 million households, either by pickling them in houses or buying them from the market.
- After analyzing the size of the local market and the volume of demand for pickles based on individual consumption of pickles, that issued by the Department of Statistics in 2013, the annual consumption was (1,768,229) kg and the annual consumption per capita was (0.26) kg.

The following table shows the annual local consumption of pickles for the years 2013-2015:

Table 3 Local Consumption of Pickles

	2013	2014	2015
population	6,530,000	6,675,000	**9,531,712**
Annual consumption (kg)	1,768,229	1,735,500	**2,478,245**
% Consumption / population	**%26**	**%26**	**%26**

Department of Statistics - Trade Statistics

The statistics of the Department of Statistics indicate the quantity of imports and exports of cucumbers, olives and others during the years 2013-2015 as shown in the following table

Table4 Quantity of Imported Pickles for The Years (2013-2015)

Type / ton	2013	2014	2015
Cucumber and granular choice, prepared or stored with vinegar or acid	375	500	261
Vegetables, fruit and other edible parts of plants prepared or preserved by vinegar or acetic acid	1,770	1,860	1060
Total	**2,045**	**2,360**	**1321**

National Exports

The following table shows national exports of pickles for the years (2013-2015):

Table5 National Exports of Pickles for The Years (2013-2015)

Type / ton	2013	2014	2015
Cucumber and granular choice, prepared or stored with vinegar or acid	40	100	275
Vegetables, fruit and other edible parts of plants prepared or preserved by vinegar or acetic acid	35	670	993
Total	**75**	**770**	**1,268**

Due to the increase in the population of Jordan to 9,814,959, the local consumption of 2016 of pickles is estimated at 2,551,889.34 kg / year assuming the same percentage, as the demand and the consumption of pickles increases as the population increases.

1.1 Analysis of Competitors

Most of the Kingdom's imports of pickles are from Syria and Turkey. However, the bulk of Jordan's consumption of them, especially olives, is covered by local production. The local market is open to the manufacture of pickles, especially cucumbers and others to replace the imported goods. The project is not expected to face difficulties in marketing its products if the production is of good quality and sold at competitive prices.

According to the Jordan Commercial Directory, there are currently in Jordan 10 factories for the production of pickles distributed in different governorates, none of them in Al Mafraq. Jordan dependeds on imports from Syria and Turkey, but due to the current situation, the market relies on local products mainly with the export of small quantities.

Such as:

1- Al Kaseeh
2- Farm dairy
3- Aden for water and juices
4- Al Mutakamelah for olive products
5- Modern Jordanian Food Industries
6- Al Khalij Food
7- Al Reiadah for food industries
8- Durra
9- Al Baraka trees for food industry
10- Dar Al Salam

Market Share

The project's share of the market can be determined based on the proposed annual capacity. The project will start producing nearly 400 tons per year, which constitutes approximately 15% of the annual consumption for 2016. The production capacity of such projects depends on laborforce and their skill rather than on equipment as well as the availability of raw materials. It is believed that the assumed employment and equipment commensurate with the estimated power.

Price Analysis and Pricing Policy

The average selling prices in local markets for 2017 for each type of locally produced pickles that were estimated for the proposed project products. The following table shows the proposed prices at an annual rate of increase of 3%:

Table 6 Prices Assumptions

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Pickles price	0.850	0.880	0.910	0.940	0.970	1.000	1.030	1.060	1.090	1.120
pickled cucumber Price	1.000	1.030	1.060	1.090	1.120	1.150	1.180	1.210	1.240	1.270
Price of pickled green olives	1.550	1.580	1.610	1.640	1.670	1.700	1.730	1.760	1.790	1.820
Price of pickled black olives	1.850	1.880	1.910	1.940	1.970	2.000	2.030	2.060	2.090	2.120
The price of pickled hot pepper	1.500	1.530	1.560	1.590	1.620	1.650	1.680	1.710	1.740	1.770

Table 7 Prices Rates

Prices Rates	
pickled cucumber Price	20%
Pickles price	20%
Price of pickled green olives	20%
Price of pickled black olives	20%
The price of pickled hot pepper	20%

Projected Revenues

Based on the total annual production capacity proposed and the annual output ratio of each product category and based on the proposed selling prices, the expected revenues of the project:

Table 8 Projected Revenues

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
pickled cucumber Price	1.000	1.030	1.060	1.090	1.120	1.150	1.180	1.210	1.240	1.270
Pickles price	0.850	0.880	0.910	0.940	0.970	1.000	1.030	1.060	1.090	1.120
Price of pickled green olives	1.550	1.580	1.610	1.640	1.670	1.700	1.730	1.760	1.790	1.820
Price of pickled black olives	1.850	1.880	1.910	1.940	1.970	2.000	2.030	2.060	2.090	2.120
The price of pickled hot pepper	1.500	1.530	1.560	1.590	1.620	1.650	1.680	1.710	1.740	1.770
Revenues	**540000**	**552000**	**564000**	**576000**	**588000**	**600000**	**612000**	**624000**	**636000**	**648000**

3. Technecial Analysis

The pickle is one of the most wanted and required appetizers in the Kingdom. The process of pickling is considered to be a process to impart taste and raise the nutritional value of vegetables and make it easy to digest.

The agricultural area in AlMafraq governorate has a high potential. It has a large area of cultivated land with an area of about 481,352 thousand dunums. Water resources are available in Al Mafraq governorate, which is rich in artesian wells.

The total area of vegetables production in AlMafraq is 68325.8 dunums. The number of greenhouses in AlMafraq is 3806.

(Source: Ministry of Agriculture statistics)

Table 9 Required Raw Materials

Product	Quantity
Cucumber	135 tons
Salt	17 tons
Water	135 cubic meters
Vinegar	250 kg

Required Human Resources

Table 10 Required Labor for The Project

Job Title	No.
Plant Manager	1
Managing Director	1
marketing	1
Accountant	2
Guard	1
Cleaner	1
Total	**7**
Production Supervisor	1
Driver	2
Production workers	8
Total	**11**

Job Description:

Factory Manager: will perform all the tasks and responsibilities of the factory manager, and have active participation in the identification and formulation of objectives, as well as planning and organizing workflow of the facility to ensure the achievement of objectives, and alignment with professional safety and security regulations.

Management Manager: Works within the authority and participates in the policy of the administrative institution and its strategic plans, planning, organization and follow-up of administrative services and control the internal management of the plant, and development of human resources to ensure the achievement of the objectives set, and the preparation of technical reports and the management of subordinates and develop their skills.

Marketing Manager: Preparing for receiving the product, testing and matching its quantities, types and specifications with the purchase order, inspecting its validity for human consumption, filling the forms of the receipt and delivery process, follow-up of the loading and unloading process, sorting the packages of the product according to type, help customers choose their needs, respond to their inquiries, secure their applications, and follow up their processing.

Production Officer: responsible for the technical, artistic and production matters and oversees the production process.

Production Staff: will work on production lines.

Accountant: will be responsible for the implementation of financial and accounting procedures in the plant.

Secretary: will respond to calls and arrange meetings for managers.

Hygiene staff: will maintain the cleanliness of the plant and offices.

Site Analysis

The location of the proposed project site is at King Hussein Bin Talal Development Area in AlMafraq Governorate to provide advantages to investors.

• Located in the heart of the Middle East, making it easier for investors to access the growing economic markets in the region.

• Located near Saudi Arabia and on the main road which connects southern Jordan to Iraq, allowing investors to benefit from their large markets.

The Production Process

The process of pickling the vegetables, including olives, begins by washing the fruits after the process of grading and sorting and then picking the fruits by placing them in the salt solution, which is prepared in advance with a concentration of 10%. As for the olives are dimple before starting the process of pickling, then packaged in containers and kept in a warm place (20-30 ° C) to speed up the pickling, and the pickling time is between 10-15 days.

It is important to note that the method of removing bitterness or reducing the bitterness of olives is done using alkaline materials, such as sodium hydroxide without the need for mechanisms of knocking or slicing the fruit by soaking the olives in an alkaline solution of 2% for 3-4 hours after the fruits are washed from the effects of the alkaline solution, and then the process of pickling as in the case of keeping the olives.

The project will manufacture the pickles in a variety forms by sorting the vegetables and washing them, then drying the vegetables and preparing them by type and then mixing the ingredients and additives and putting them in brine to save them from corruption, then finishing the process of pickling by filling and packaging the products in special containers of different capacities.

Technical Requirements Analysis

1.1.1 Land and Construction

The project will be constructed on an area of 2000 sq.m. The following table shows the needed land:

Table11 Required Land for The Project

Land	Area sq.m
The required land	**2000**

Source: Land registration – Al Mafraq

The following table shows the required construction work for the project:

Table 12 Required Construction Work

Construction Work	Area (m²)
Hangars	1,400
Settling	1,400
Paving	600
Fences	200

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 13 Machinery and equipments

Machinery and Equipment	Quantity
Olive Processing Line	1
Olive grove	1
Punching Machine	1
Sorting machine	1
Pepper Grinder	1
Filling machine	2
Dock for mixing ingredients	2
Generator	1
Cutting machine for carrots	1
Drums of food plastic	1000

Vehicles and Transportation

The following table shows a list of vehicles and means of transportion required for the project

Table 14 Vehicles and Transportation

Vehicles and Transportation	Quantity	Unit cost (JD)
Pickup truck	2	23,400

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.61% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 15 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 16 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 18 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	40%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	1%
Depreciation Expenses from Total Revenues	1.0%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	1%

Table 19 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	**Value**
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premuim	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premuim (3)	10.12%
Income Tax Rate	5%
Beta (4)	0.89
Risk Premuim	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 20 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%5
Bets	0.89
Equity	268,106
Loans Value	402,159
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	402,159
Loans Percentage	%60
Equity	
Cost of Equity	%15.5
Equity Value	268,106
Equety Percentage	%40
Gross	
Project Value	670,264
Proportion of the project	%100
Weighted Average Cost of Capital	%11.61

Capital Expenditures

The estimated cost of the project is 670,264 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 21 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	25	2,000	50,000
Cost			**50,000**

Table 22 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Henger	100	1,400	140,000
Settling	15	1,400	21,000
Paving	30	600	18,000
Fences	80	200	16,000
Total			**195,000**

Table 23 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Olive Processing Line	1	4,000	4,000
Olive grove	1	2,800	2,800
خزافة خيار	1	4,000	4,000
Sorting machine	1	85,000	85,000
Pepper Grinder	1	2,000	2,000
Filling machine	2	3,500	7,000
Dock for mixing ingredients	2	500	1,000
Generator	1	23,000	23,000
Cutting machine for carrots	1	3,500	3,500
Drums of food plastic	1000	60	60,000
Total (JD)			**200,300**

Table 24 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	12,000
Total			12,000

Table 25 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,400	46,800
Total			**46,800**

Table 26 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,000
Transportation	750
Marketing	1,500
Training	1,000
Total	**4250**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 27 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	78,250
General and administrative expenses	3	%25	2,381
Marketing expenses	3	%25	1,350
Indirect Salaries	3	%25	19,000
Total			**100,981**

Table 28 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	50,000	10%	55,000	%8.2
Construction works	195,000	10%	214,500	%32.0
Machinery &Equipment	200,300	10%	220,330	%32.9
Furniture	12,000	10%	13,200	%2.0
Vehicles	46,800	10%	51,480	%7.7
Pre-Operating Expenses	4,250	10%	4,675	%0.7
Working capital	100,981	10%	111,079	%16.6
Total (JD)	**609,331**		**670,264**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (670,264) JD, where 60% of the project will be financed through loans (402,159) JD, and the remaining (40%) through (Equity) with (268,106) JD. The following table summarizes the general structure of the required financing:

Table 29 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	268,106	%40
Loans	402,159	%60
Total	**670,264**	**%100**
Use of Fund	Value	Ratio
Capital Expenditures	554,510	%82.73
Pre-Operating Expenses	4,675	%0.70
Working Capital	111,079	%16.57
Total	**670,264**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 30 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	216,000	220,800	225,600	230,400	235,200	240,000	244,800	249,600	254,400	259,200
Utilities from	27,000	27,600	28,200	28,800	29,400	30,000	30,600	31,200	31,800	32,400
Maintenance from	5,400	5,520	5,640	5,760	5,880	6,000	6,120	6,240	6,360	6,480
Disposables from	5,400	5,520	5,640	5,760	5,880	6,000	6,120	6,240	6,360	6,480
Direct Salaries	59,200	61,627	64,154	66,784	69,522	72,373	75,340	78,429	81,645	84,992
Others	31,300	32,107	32,923	33,750	34,588	35,437	36,298	37,171	38,056	38,955
Total	**313,000**	**321,067**	**329,234**	**337,504**	**345,882**	**354,373**	**362,980**	**371,709**	**380,565**	**389,552**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 31 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	500	515	530	546	563	580	597	615	633	652

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Staionary and Printing	500	515	530	546	563	580	597	615	633	652
Transportaton	1,800	1,854	1,910	1,967	2,026	2,087	2,149	2,214	2,280	2,349
Training	500	515	530	546	563	580	597	615	633	652
Insurance	4,159	4,284	4,412	4,544	4,681	4,821	4,966	5,115	5,268	5,426
Miscellaneous	866	892	919	946	975	1,004	1,034	1,065	1,097	1,130
Total	**9,525**	**9,810**	**10,105**	**10,408**	**10,720**	**11,042**	**11,373**	**11,714**	**12,066**	**12,428**

Marketing Expenses

Table 32 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**5,400**	**5,520**	**5,640**	**5,760**	**5,880**	**6,000**	**6,120**	**6,240**	**6,360**	**6,480**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 33 Expected staff preparation

Job title	No.	Monthly salaries	Total monthly salaries	Total yearly salaries
Plant Manager	1	1,500	1,500	24,000
Accountant	2	450	900	14,400
Guard	1	250	250	4,000
Managing Director	1	1,400	1,400	22,400
marketing	1	450	450	7,200
Cleaner	1	250	250	4,000
Total Indirect staff	**7**		**4,750**	**76,000**
Production Supervisor	1	800	800	12,800
Driver	2	250	500	8,000
Production workers	8	300	2,400	38,400
Total direct staff	11		3,700	59,200
Grand Total	**18**		**8,450**	**135,200**

Table 34 Expected monthly salaries (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Plant Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Accountant	450	468	488	508	528	550	573	596	621	646
Guard	250	260	271	282	294	306	318	331	345	359

Managing Director	1,400	1,457	1,517	1,579	1,644	1,712	1,782	1,855	1,931	2,010
marketing	450	468	488	508	528	550	573	596	621	646
Cleaner	250	260	271	282	294	306	318	331	345	359
Direct staff										
Production Supervisor	800	833	867	902	939	978	1,018	1,060	1,103	1,149
Driver	250	260	271	282	294	306	318	331	345	359
Production workers	300	312	325	338	352	367	382	397	414	431

Table 31 Annual Expected Salaries

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Plant Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Accountant	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Guard	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Managing Director	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159
marketing	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Cleaner	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Total Indirect	**76,000**	**79,116**	**82,360**	**85,737**	**89,252**	**92,911**	**96,720**	**100,686**	**104,814**	**109,111**
Direct staff										
Production Supervisor	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Driver	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Production workers	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Total Direct	**59,200**	**61,627**	**64,154**	**66,784**	**69,522**	**72,373**	**75,340**	**78,429**	**81,645**	**84,992**
Grand Total	**135,200**	**140,743**	**146,514**	**152,521**	**158,774**	**165,284**	**172,060**	**179,115**	**186,459**	**194,103**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 35 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	55,000	%0.0	%0.0
Construction Works	214,500	%5.0	%0.0
Machinaries	220,330	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0
	554,510		

Table 36 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	554,510	559,577	564,764	570,077	575,517	581,089	586,797	592,644	598,634	604,772
Total Depreciation Expenses	39,226	39,733	40,251	40,783	41,327	41,884	42,455	43,039	43,638	44,252

Total Accumulated Depreciation	39,226	78,959	119,210	159,993	201,319	243,203	285,658	328,697	372,336	416,588
Total Additions	0	5,067	5,188	5,312	5,440	5,572	5,708	5,847	5,990	6,138
Total Net Book Values	515,284	480,618	445,554	410,084	374,198	337,886	301,139	263,946	226,298	188,184

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 37 Loan Details

Loan Value	402,159
Annual Interst Rate	9.48%
Loan Starts at year:	2018
Annual Payment	104,683
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				402,159
2019	104,683	38,125	66,558	335,600
2020	104,683	31,815	72,868	262,732
2021	104,683	24,907	79,776	182,957
2022	104,683	17,344	87,339	95,618

2023	104,683	9,065	95,618	0

Income Statement

Table 38 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	540,000	552,000	564,000	576,000	588,000	600,000	612,000	624,000	636,000	648,000
Gross Operating Revenues	540,000	552,000	564,000	576,000	588,000	600,000	612,000	624,000	636,000	648,000
Operating Expenses	-313,000	-321,067	-329,234	-337,504	-345,882	-354,373	-362,980	-371,709	-380,565	-389,552
Gross Operating Profit	227,000	230,933	234,766	238,496	242,118	245,627	249,020	252,291	255,435	258,448
Gross Profit Percentage	*%42*	*%42*	*%42*	*%41*	*%41*	*%41*	*%41*	*%40*	*%40*	*%40*
Salaries and Benefits (Indirect Staff)	-76,000	-79,116	-82,360	-85,737	-89,252	-92,911	-96,720	-100,686	-104,814	-109,111
General and Administraive Expenses	-9,525	-9,810	-10,105	-10,408	-10,720	-11,042	-11,373	-11,714	-12,066	-12,428
Markeing Expenses	-5,400	-5,520	-5,640	-5,760	-5,880	-6,000	-6,120	-6,240	-6,360	-6,480
Pre Operating Expenses	-4,675									
Gross Indirect Expenses	-95,600	-94,446	-98,105	-101,904	-105,852	-109,953	-114,213	-118,640	-123,240	-128,019
Income before Interest, Depreciation and Tax	131,400	136,486	136,662	136,591	136,266	135,674	134,807	133,651	132,196	130,429

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets Depreciations	-39,226	-39,733	-40,251	-40,783	-41,327	-41,884	-42,455	-43,039	-43,638	-44,252
Income before Tax and Interests	92,174	96,754	96,410	95,809	94,939	93,791	92,352	90,612	88,557	86,177
Bank Interests	-38,125	-31,815	-24,907	-17,344	-9,065	-0	-0	-0	-0	-0
Income Before Tax	54,050	64,939	71,503	78,464	85,874	93,791	92,352	90,612	88,557	86,177
Income Tax	-2,702	-3,247	-3,575	-3,923	-4,294	-4,690	-4,618	-4,531	-4,428	-4,309
Net Profit	51,347	61,692	67,928	74,541	81,581	89,101	87,734	86,081	84,129	81,868
Net Profit Percentage	*%10*	*%11*	*%12*	*%13*	*%14*	*%15*	*%14*	*%14*	*%13*	*%13*
Compulsory Reserves	-5,135	-6,169	-6,793	-7,454	-8,158	-8,910	-8,773	-8,608	-8,413	-8,187
Retained Earnings	46,212	101,735	162,870	229,957	303,380	383,571	462,532	540,005	615,721	689,402

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 39 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	51,347	61,692	67,928	74,541	81,581	89,101	87,734	86,081	84,129	81,868
Non-monetary adjustments										
Bank benefits	38,125	31,815	24,907	17,344	9,065	0	0	0	0	0

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Consumption	39,226	39,733	40,251	40,783	41,327	41,884	42,455	43,039	43,638	44,252
Operating cash flows before changes in working capital	128,698	133,239	133,086	132,668	131,972	130,985	130,189	129,120	127,768	126,120
Shortage (increase) in inventory	-18,000	-400	-400	-400	-400	-400	-400	-400	-400	-400
(Decrease) increase in accounts payable	16,167	545	561	578	596	615	635	655	676	698
Change in working capital	-1,833	145	161	178	196	215	235	255	276	298
Net cash flows (used) from operating activities	126,864	133,384	133,248	132,847	132,168	131,200	130,423	129,375	128,044	126,418
Cash flows from investing activities										
(Purchase) of fixed assets	-554,510	-5,067	-5,188	-5,312	-5,440	-5,572	-5,708	-5,847	-5,990	-6,138
Net flows (used) from investment activities	-554,510	-5,067	-5,188	-5,312	-5,440	-5,572	-5,708	-5,847	-5,990	-6,138
Cash flows from financing activities										
Equity capital	268,106									
Loan Amortization	-66,558	-72,868	-79,776	-87,339	-95,618	0	0	0	0	0
Bank Interest Rate	-38,125	-31,815	-24,907	-17,344	-9,065	-0	-0	-0	-0	-0
Loans	402,159	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	565,581	-104,683	-104,683	-104,683	-104,683	0	0	0	0	0
Net increase (decrease) in cash	137,936	23,635	23,377	22,851	22,045	125,628	124,716	123,528	122,053	120,280
Cashflows at the Beginning of Period	0	137,936	161,570	184,947	207,799	229,844	355,472	480,188	603,716	725,769
Cashflows at the End of Period	137,936	161,570	184,947	207,799	229,844	355,472	480,188	603,716	725,769	846,049

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 40 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	137,936	161,570	184,947	207,799	229,844	355,472	480,188	603,716	725,769	846,049
Inventory	18,000	18,400	18,800	19,200	19,600	20,000	20,400	20,800	21,200	21,600
Total assets traded	**155,936**	**179,970**	**203,747**	**226,999**	**249,444**	**375,472**	**500,588**	**624,516**	**746,969**	**867,649**
Non-current assets										
Fixed assets (net)	515,284	480,618	445,554	410,084	374,198	337,886	301,139	263,946	226,298	188,184
Total non-current assets	**515,284**	**480,618**	**445,554**	**410,084**	**374,198**	**337,886**	**301,139**	**263,946**	**226,298**	**188,184**
Total assets	**671,220**	**660,588**	**649,302**	**637,083**	**623,642**	**713,358**	**801,727**	**888,462**	**973,268**	**1,055,834**
Liabilities	-	-	-	-	-					
Current liabilities										
Payables	16,167	16,711	17,272	17,851	18,447	19,062	19,697	20,352	21,027	21,725
Remaining amount of Loan	72,868	79,776	87,339	95,618	0-	0-	0-	0-	0-	0
Total current liabilities	**89,035**	**96,487**	**104,611**	**113,469**	**18,447**	**19,062**	**19,697**	**20,352**	**21,027**	**21,725**

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	262,732	182,957	95,618	0	0	0	0	0	0	0
Total long - term liabilities	**262,732**	**182,957**	**95,618**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total liabilities	**351,767**	**279,444**	**200,229**	**113,469**	**18,447**	**19,062**	**19,697**	**20,352**	**21,027**	**21,725**
Property rights										
Member Contributions	268,106	268,106	268,106	268,106	268,106	268,106	268,106	268,106	268,106	268,106
Compulsory reserve	5,135	11,304	18,097	25,551	33,709	42,619	51,392	60,001	68,413	76,600
Profit (loss) round	46,212	101,735	162,870	229,957	303,380	383,571	462,532	540,005	615,721	689,402
Total equity	319,453	381,145	449,073	523,614	605,195	694,296	782,030	868,111	952,240	1,034,108
Total liabilities and equity	**671,220**	**660,588**	**649,302**	**637,083**	**623,642**	**713,358**	**801,727**	**888,462**	**973,268**	**1,055,834**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 41 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-670,264	131,539	128,317	128,060	127,534	126,728	125,628	124,716	123,528	122,053	120,280	1,644,568
Discount Factor	1.00	0.80	0.72	0.64	0.58	0.52	0.46	0.42	0.37	0.33	0.30	0.30

Net Present Value for Cash	**-670,264**	**105,604**	**92,304**	**82,539**	**73,652**	**65,575**	**58,246**	**51,810**	**45,980**	**40,706**	**35,943**	**491,445**

Table below (42) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 42 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-670,264	131,539	128,317	128,060	127,534	126,728	125,628	124,716	123,528	122,053	120,280
Project Value Returned	670,264	538,725	410,407	282,348	154,813	28,085	-97,543	-222,258	-345,787	-467,840	-588,120
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 43 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%11.61
Net Present Value for Cashflows	473,539
Payback Period	4
Internal Rate of Return	%22.45

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 473,539 Jordan Dinars considering that the project provides 18 Job Opportunitites for the governorate residents.

Table 44 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%22.45	4	11.61%
Revenues declined by 10%	%17.33	4	11.61%
Operating Expenses Increased by 10%	%16.27	4	11.61%